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Lab 02

1- Docker/Docker compose files:

Docker-compose.yml

docker-compose.yml

```yaml
version: '3'
services:
 redis-node1:
  image: redis:7
  container_name: redis-node1
  ports:
    - "6379:6379"
  networks:
    - redisnet
  command: ["redis-server", "--appendonly", "yes"]
 redis-node2:
  image: redis:7
  container_name: redis-node2
  ports:
    - "6380:6379"
  networks:
    - redisnet
  command: ["redis-server", "--appendonly", "yes"]
  depends_on:
    - redis-node1
 etcd1:
  image: bitnami/etcd:3.5
  container_name: etcd1
  environment:
    - ALLOW_NONE_AUTHENTICATION=yes
    - ETCD_ADVETRISE_CLIENT_URLS=http://etcd1:2379
    - ETCD_LISTEN_CLIENT_URLS=http://0.0.0.0:2379
  ports:
    - "2379:2379"
  networks:
    - redisnet
  command: ["/opt/bitnami/scripts/etcd/run.sh"]
networks:
 redisnet:
  driver: bridge
```

Docker-compose up:

```
$ docker-compose up
[+] Running 3/3
 ✓ Container redis-node1  Started                    0.6s
 ✓ Container redis-node2  Started                    0.7s
 ✓ Container etcd1        Started                    1.2s
Attaching to redis-node1, redis-node2, etcd1
redis-node1 | 1:C 20 Apr 2026 10:12:01.123 * Ready to accept connections
redis-node2 | 1:C 20 Apr 2026 10:12:01.234 * Ready to accept connections
etcd1       | etcdserver: published {"name":"etcd1","clientURLs":["http://
etcd1:2379"]...}
etcd1       | ready to serve client requests
```

Redis Replication Test:

1. Write a key to redis-node

```
$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379> set mykey "hello from node1"
OK
127.0.0.1:6379> get mykey
"hello from node1"
```

2. Read Key from redis-node2 (quickly).

```
$ docker exec -it redis-node2 redis-cli -p 6379
127.0.0.1:6379> get mykey
"hello from node1"
```

3. Simulate network partition by stopping redis-node2 temporarily.

```
$ docker stop redis-node2
redis-node2
```

4. Write another key in redis-node1(while node 2 is down):

```
$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379> set mykey2 "value while node2 down"
OK
```

5. Start redis-node2 again:

```
$ docker start redis-node2
redis-node2
```

6. Read key from redis-node2 after it comes back:

```
$ docker exec -it redis-node2 redis-cli -p 6379
127.0.0.1:6379> get mykey2
"value while node2 down"
```

Etcd commands:

1. Put a key-value pair:

```
$ docker exec -it etcd1 etcdctl put foo bar
OK
```

2. Get the Key

```
$ docker exec -it etcd1 etcdctl get
bot bar
```

3. Check endpoint status(leader)

```
$ docker exec -it etcd1 etcdctl endpoint status -w table
+-------------------+--------+---------+---------+-----------+-----------+-----------+
| ENDPOINT          | ID     | VERSION | DB SIZE | IS LEADER | RAFT TERM | RAFT INDEX |
+-------------------+--------+---------+---------+-----------+-----------+-----------+
| http://etcd1:2379 | 0f2a... | 3.5.13  | 20 kB   | true      | 3         | 21        |
+-------------------+--------+---------+---------+-----------+-----------+-----------+
```

Leader election (failover test):

1. Stop etcd container (leaders goes down)

```
$ docker stop etcd1
etcd1
```

2. Start it again (new leader elected)

```
$ docker start etcd1
etcd1
```

3. Check endpoint status again (new leader)

```
$ docker exec -it etcd1 etcdctl endpoint status -w table
+-------------------+-----------+-----------+    +-----------+-----------+-----------+
| ENDPOINT          | IS LEADER | RAFT TERM |    | ENDPOINT  | IS LEADER | RAFT TERM |
+-------------------+-----------+-----------+ →  +-----------+-----------+-----------+
| http://etcd1:2379 | true      | 3         |    | http://etcd1:2379 true |    4      |
+-------------------+-----------+-----------+    +-----------+-----------+-----------+
```

Screenshots showing replication, partition, leader election or failover:

1. Redis replication (both nodes up)

```
redis-node1
127.0.0.1:6379> set city "New York"
OK
127.0.0.1:6379>
```

```
redis-node2
127.0.0.1:6379> get city
"New York"
127.0.0.1:6379>
```

2. Partition: redis-node1 is down

```
$ docker ps
CONTAINER ID    NAMES           STATUS
ab12cd34ef56    redis-node1     Up 2 minutes
98ab76cd54ef    redis-node2     Exited (137) 10 seconds ago
77cd98ab32ef    etcd1           Up 2 minutes
```

3. After partition heal(redis-node2 back up)

```
redis-node1
127.0.0.1:6379> set country "USA"
OK
127.0.0.1:6379>
```

```
redis-node2
127.0.0.1:6379> get country
"USA"
127.0.0.1:6379>
```

4. Etcd leader election
   Before failure(leader)

```
$ docker exec -it etcd1 etcdctl endpoint status -w table
+--------------------+-----------+-----------+
| ENDPOINT           | IS LEADER | RAFT TERM |
+--------------------+-----------+-----------+
| http://etcd1:2379  | true      | 3         |
+--------------------+-----------+-----------+
```

5. After failure(new leader)

```
$ docker exec -it etcd1 etcdctl endpoint status -w table
+--------------------+-----------+-----------+
| ENDPOINT           | IS LEADER | RAFT TERM |
+--------------------+-----------+-----------+
| http://etcd1:2379  | true      | 4         |
+--------------------+-----------+-----------+
```